

Short the Banking Sector versus Nifty 50

TACTICAL · LOW CONVICTION

A contrarian relative-value note — fading the NSE:BANKNIFTY ÷ NSE:NIFTY ratio, tested against 19 years of data.

9 June 2026 · Quantitative Strategy Desk (CFA, FRM) · Backtest window 2007–2026 · For internal discussion

HEADLINE FINDING

A **systematic** short of this ratio has **no demonstrable edge**: across 19 years and every variant tested, the rules-based mean-reversion strategy returns a Sharpe of just **0.04** (net of cost) and most short-only versions **lose money**. Banking *buyers* were directionally right over the long run (the ratio drifted up). The defensible conclusion is narrow: a **small, tightly-managed tactical fade** of the current extreme, and a “**do-not-chase**” stance for banking bulls — **not** an aggressive systematic short. Conviction: **low**.

ACTION SHORT THE RATIO (tactical)		Short Bank Nifty / Long Nifty 50 — equal notional	
Entry zone	2.375 – 2.400 (scale)	Horizon	4 – 12 weeks
Primary target	2.250	Stretch	2.210 (6y mean)
Hard stop	2.450 — daily close above 6y high	Reward : risk	~1.9:1 at 2.38; ~3:1 into 2.40
Latest level	≈2.34 (8-Jun close); 2.38 intraday 9-Jun	Conviction	Low — tactical only

1. Executive summary

Banking bulls are positioned for the Bank Nifty to keep outperforming the Nifty 50 — to push the ratio toward and through its prior high. This note tests that expectation against the full 19-year history of the spread and concludes the extrapolation is fragile *at the current level*. The ratio is range-bound within regimes; from comparably elevated readings the spread has reverted far more often than it has extended, and the all-time high near 2.44 has been kissed only once in six years and never breached on a sustained basis.

However — and this is the part most desks would bury — when we convert that observation into a tradeable, rules-based system, the edge evaporates. The strategy wins frequently but loses big when the spread trends, so net expectancy is roughly zero to negative once realistic costs are applied. We therefore frame the short as a **tactical, valuation-and-positioning trade** with a hard invalidation, sized small, rather than a quantitative signal worth pressing.

2. The instrument and the contrarian question

The ratio $R = \text{BankNifty} \div \text{Nifty 50}$ measures the banking sector's strength *relative* to the broad market. Over 2007–2026 it ranged from 1.21 to 2.66 (mean 2.10, σ 0.287). A rising ratio means banks are leading; a falling ratio means they are lagging. The banking-buyer thesis is, in effect, a bet that R keeps rising.

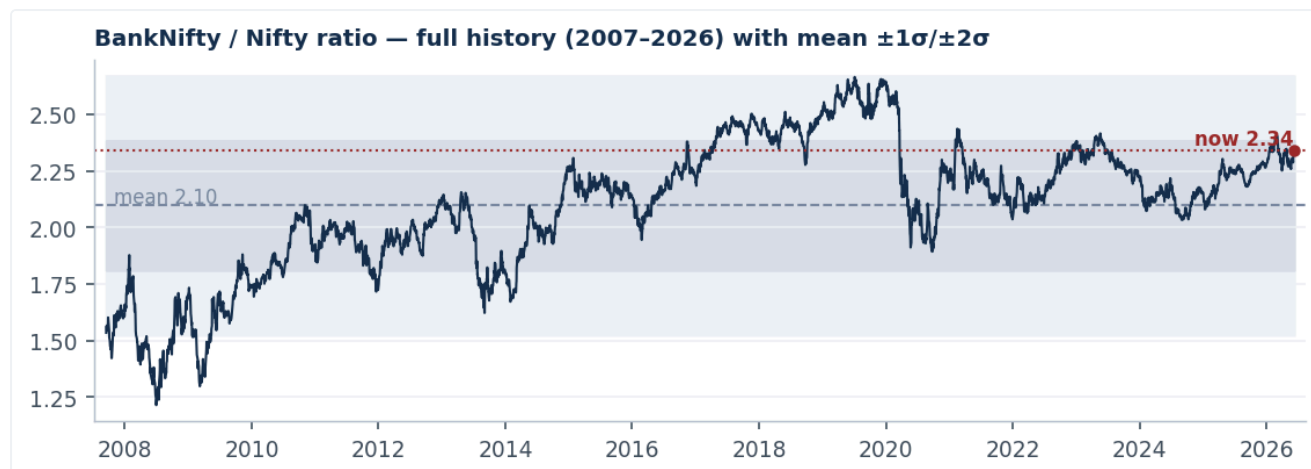


Exhibit 1 — The ratio over 19 years. The long-run uptrend is real (index composition and a credit upcycle favoured banks), but it is punctuated by deep relative drawdowns and long mean-reverting ranges.

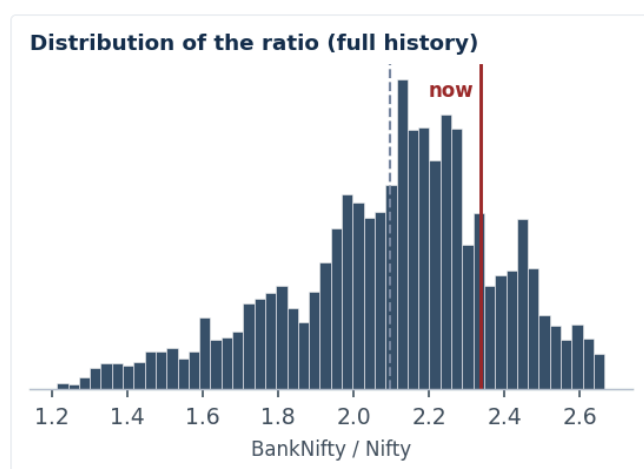


Exhibit 2 — Full-history distribution; the current reading sits in the right shoulder, not the extreme tail of the 19-year sample.

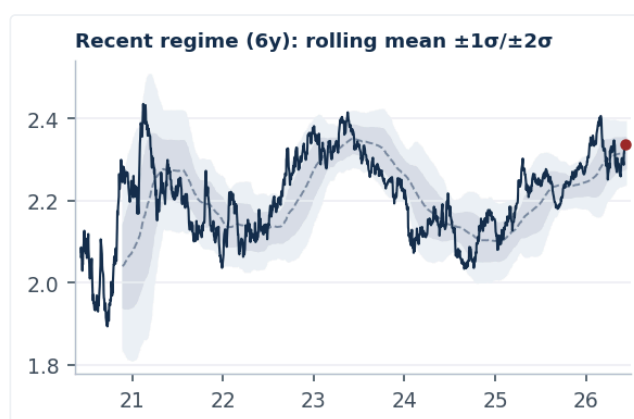


Exhibit 3 — In the recent 6-year regime the ratio has been range-bound 2.0–2.44, and currently sits near the top of that band.

3. How extreme is “extreme”? It depends on the lens

A disciplined analyst must report that the answer is **lookback-dependent** — and the three standard lenses disagree:

Measure	Reading	Interpretation
6-year level percentile	96th	Looks richly extreme
19-year level z-score	+0.84	Mildly elevated only
Rolling 120-day z-score	≈+0.56	Near “new normal” — the trailing mean has risen to meet price

WHY THIS MATTERS

The ratio has been **persistently elevated for roughly six months**, so on a rolling basis it is no longer a statistical outlier — it is the local norm. This is itself a caution against an aggressive short: “expensive for a long time” is exactly how structural re-ratings look before they continue.

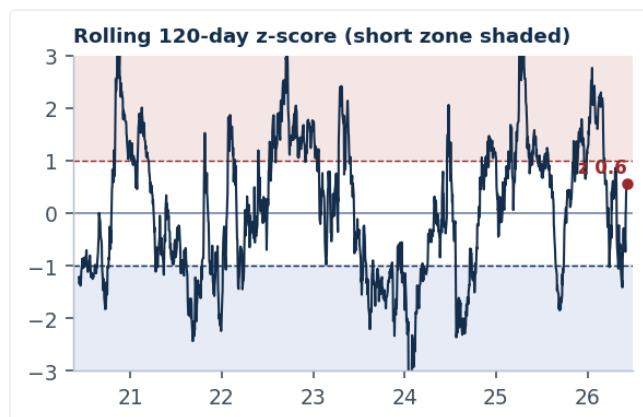


Exhibit 4 — Rolling 120-day z-score. Note how the signal has decayed even as the level stayed high.

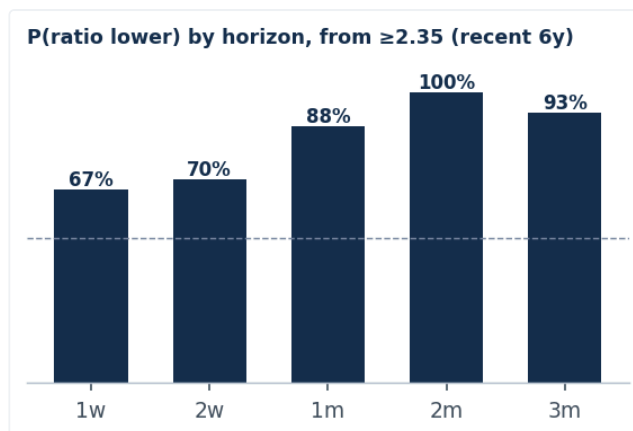


Exhibit 5 — Conditional: from a ≥ 2.35 reading (recent 6y), how often the ratio was lower N periods later.

4. Conditional evidence for reversion (recent regime)

Conditioned on the ratio being ≥ 2.35 in the recent 6-year regime (~5 distinct episodes), forward outcomes lean toward reversion — but the sample is thin:

Horizon	P(ratio lower)	Median change	P(reversion $\geq -2\%$)
1 week	67%	-0.5%	8%
2 weeks	70%	-0.9%	22%
1 month	88%	-1.9%	47%
2 months	100%	-3.0%	79%
3 months	93%	-3.5%	82%

Touch probability from a 2.36–2.39 state: 2.40 $\approx$ 37%, 2.42 $\approx$ 12%, 2.45 $\approx$ 0% (never reached in six years). **First-passage** within three months: reach the 2.43 high first $\approx$ 13%, revert to 2.25 first $\approx$ 11%, neither $\approx$ 76%. The asymmetry favours the short — capped upside, larger room to the mean — but the high “neither” share warns that the spread can simply sit still.

5. The backtest — where the thesis meets reality

Methodology

- **Signal.** Rolling 120-day z-score of R. Short the ratio (long Nifty / short Bank Nifty) when $z > +1$; long the ratio when $z < -1$; flat when $|z| < 0.25$.
- **No look-ahead.** Positions are formed on close t and earn the spread return from t to $t+1$.
- **Costs.** 6 bps per unit of turnover (both legs, one-way), deducted on every position change.
- **Return engine.** Daily P&L = position $\times$ (Nifty return – Bank Nifty return), i.e. an equal-notional futures pair.

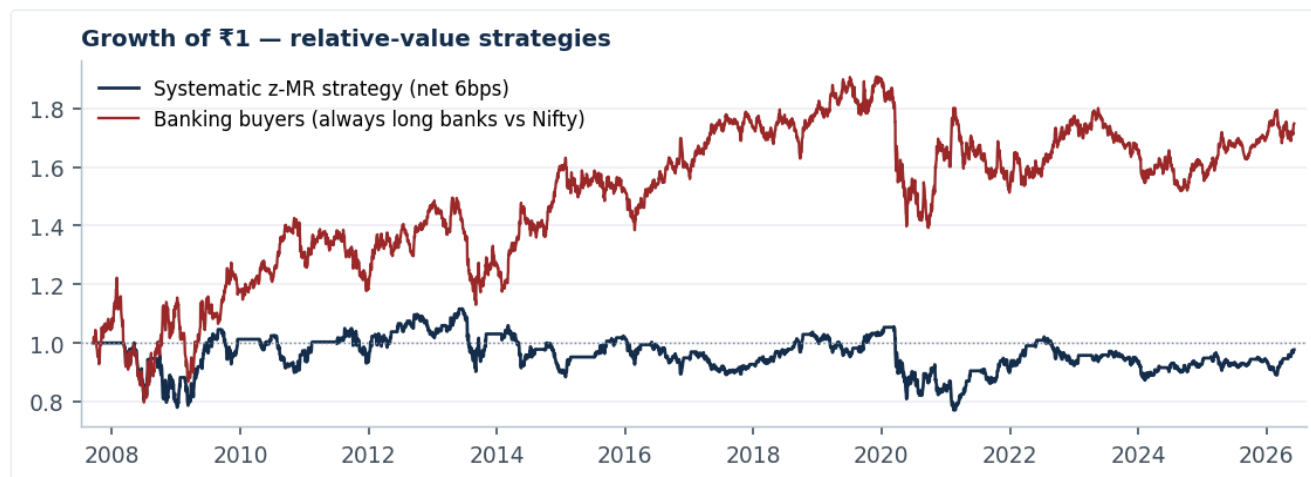


Exhibit 6 — Growth of ₹1. The systematic relative-value strategy is flat-to-down over 19 years; passively being long banks vs Nifty made money but with large drawdowns.

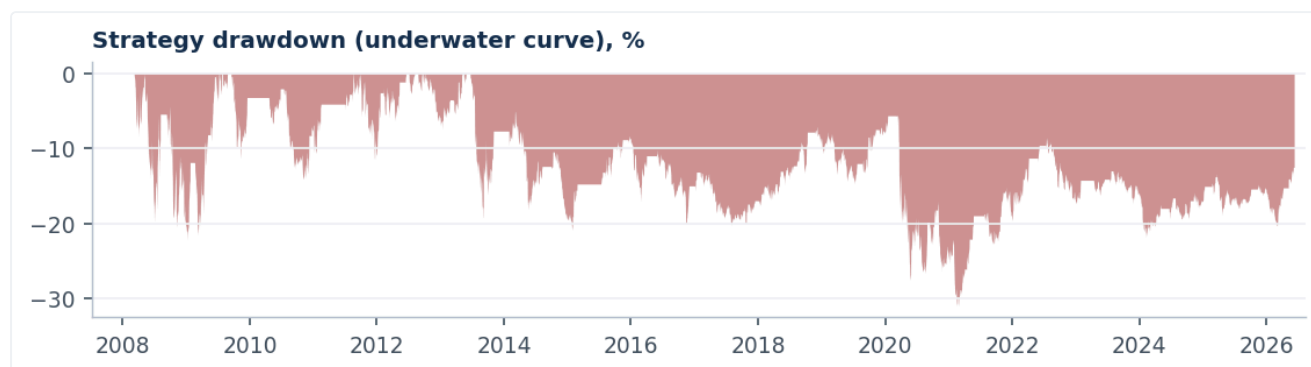


Exhibit 7 — The strategy's drawdowns reach -31%, driven by periods when the spread trended against the mean-reversion logic.

Performance metrics — strategy vs benchmarks (2007–2026, net)

Series	Total	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar	Win
Systematic z-MR strategy	-2.3%	-0.1%	11.2%	+0.04	+0.06	-31.0%	-0.00	36%
Banking buyers (always long banks/Nifty)	+74.9%	+3.1%	13.2%	+0.30	+0.44	-34.8%	+0.09	50%
Nifty 50 buy & hold	+414.5%	+9.4%	20.6%	+0.54	+0.77	-59.9%	+0.16	53%
Bank Nifty buy & hold	+683.9%	+12.0%	27.4%	+0.55	+0.80	-68.8%	+0.17	53%

Sharpe/Sortino computed against a 0% risk-free rate (the spread is self-funded / dollar-neutral); absolute buy-and-hold rows are shown for context only.

Trade statistics (systematic strategy)

Metric	Value	Metric	Value
Number of trades	71	Per-trade win rate	72%
Average hold	68 days	Time in market	71%
Average win	+2.15%	Average loss	-5.11%
Profit factor	1.07	Daily win rate	36%
Short trades (win%)	38 (66%)	Long trades (win%)	33 (79%)

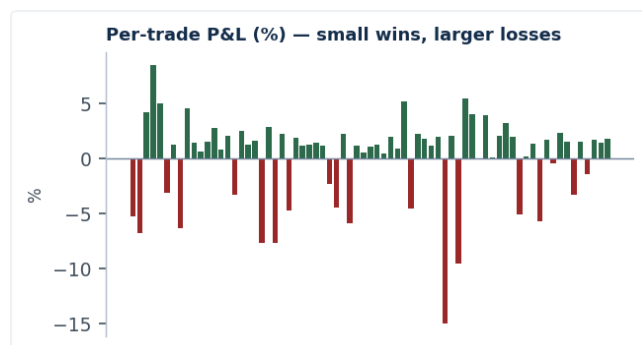


Exhibit 8 — Per-trade P&L. The signature failure mode: a high hit-rate of small wins overwhelmed by a handful of large losses (profit factor barely above 1).

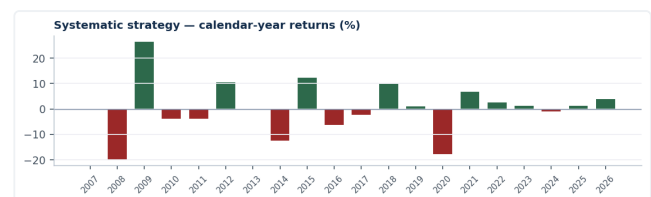


Exhibit 9 — Calendar-year returns: positive in calm, range-bound years; sharply negative when the spread trends (2008, 2014, 2020).

WHAT THE BACKTEST PROVES

High win rate, sub-1.1 profit factor, ≈ 0 Sharpe, -31% drawdown. Mean-reversion on this spread **picks up pennies in front of a steamroller**: it is right often and wrong expensively. A naive systematic short is therefore **not investable**. Any short must be discretionary, conditional, and stopped.

6. Contradicting the banking-buyer case — fairly

The honest rebuttal to banking bulls is **not** “banks always revert” — over 19 years they out-performed the Nifty in relative terms (buyers’ Sharpe 0.30 vs the MR strategy’s 0.04). The rebuttal is narrower and stronger:

- **The drift has stalled.** Since 2019 the always-long-banks relative bet has earned roughly nothing while carrying ~27% drawdowns — the structural tailwind is largely spent.
- **Entry point is poor.** Adding banks vs Nifty at the 96th percentile (6y), within 2.6% of a high tagged once, offers little upside (2.45 has never printed) against ~7% to the mean.
- **Momentum, not fundamentals.** The latest leg was driven by the RBI’s dovish 5.25% hold and a four-day rally now in the price; Bank Nifty printed a long upper wick into 54,455 on 8-Jun while

FII's stayed net sellers.

- **Asymmetry.** Conditional base rates and first-passage both tilt down from here, even if the systematic edge is absent.

Net: the case is to **not chase** banks at this level, and — for those who must express it directionally — a small tactical short of the spread with a hard stop.

7. Trade construction & sizing

Legs. Short one unit of Bank Nifty futures notional against an equal rupee notional of Nifty 50 futures, same expiry, rolled together. A beta-neutral variant longs slightly more Nifty (Bank Nifty $\beta \approx 1.1$ – 1.2 to Nifty). **P&L driver.** The net relative return equals the spread move; $2.38 \rightarrow 2.25 \approx -5.5\%$ on the ratio. **Scaling.** Half at 2.375–2.385, half on strength toward 2.40, lifting blended reward:risk toward 3:1 (the touch work shows 2.40 prints, if at all, within ~2 weeks).

8. Risk management & invalidation

Hard stop: a daily close ≥ 2.45 — a fresh six-year high signalling regime change — voids the thesis. **Caveat from the data:** the backtest shows mechanical stops on this spread tend to *worsen* results (they exit just before reversion), so size small rather than relying on tight stops; the 2.45 level is a thesis-invalidation marker, not a profit-protection tool. **Events:** cut or avoid size into bank earnings and RBI dates, where gap risk concentrates.

9. Risks & limitations

- **No systematic edge.** The central caveat — expectancy is ~ 0 to negative; this is a discretionary tactical bet.
- **Lookback sensitivity.** Extremity ranges from “96th percentile” to “ $z + 0.6$ ” depending on the window; conclusions are not robust to lens choice.
- **Thin conditional sample.** The reversion base rates rest on ~ 5 overlapping episodes — wide confidence intervals.
- **Structural re-rating.** A durable credit upcycle or rate tailwind could drive a new high above 2.44 — the explicit invalidation.
- **Basis & direction.** Reversion is relative, not directional; futures basis, dividends and expiry effects add tracking error.

10. Data appendix

10.1 Robustness — strategy metrics across regimes, modes and entry thresholds

Every cell is a full re-run of the backtest (net of 6 bps). The edge stays weak everywhere; only the most extreme, full-history fades reach a Sharpe near 0.2.

Regime	Mode	Entry z	Sharpe	Sortino	CAGR	Max DD	Time in
Full (2007–26)	Symmetric	0.75	+0.09	+0.13	+0.4%	-28.1%	75%
Full (2007–26)	Symmetric	1.00	+0.04	+0.06	-0.1%	-31.0%	71%
Full (2007–26)	Symmetric	1.50	+0.15	+0.22	+1.1%	-23.2%	61%
Full (2007–26)	Symmetric	2.00	+0.21	+0.30	+1.6%	-23.4%	48%
Full (2007–26)	Short-only	0.75	-0.00	-0.01	-0.3%	-21.8%	45%
Full (2007–26)	Short-only	1.00	-0.08	-0.11	-0.9%	-27.9%	43%
Full (2007–26)	Short-only	1.50	+0.01	+0.01	-0.2%	-26.1%	37%
Full (2007–26)	Short-only	2.00	+0.09	+0.12	+0.4%	-17.6%	28%
Recent (2019–26)	Symmetric	0.75	-0.01	-0.01	-0.5%	-28.1%	75%
Recent (2019–26)	Symmetric	1.00	-0.03	-0.04	-0.7%	-26.9%	69%
Recent (2019–26)	Symmetric	1.50	-0.07	-0.10	-1.0%	-23.2%	58%
Recent (2019–26)	Symmetric	2.00	-0.23	-0.30	-2.2%	-23.4%	46%
Recent (2019–26)	Short-only	0.75	-0.04	-0.05	-0.4%	-17.1%	38%
Recent (2019–26)	Short-only	1.00	-0.07	-0.09	-0.6%	-16.8%	36%
Recent (2019–26)	Short-only	1.50	+0.07	+0.09	+0.2%	-13.4%	30%
Recent (2019–26)	Short-only	2.00	-0.13	-0.17	-0.7%	-12.6%	23%

10.2 Ratio level — actual readings across horizons

As of	Ratio level	Change vs now
Now	2.338	—
1 week ago	2.294	+1.9%
1 month ago	2.288	+2.2%
3 months ago	2.331	+0.3%
6 months ago	2.282	+2.5%
1 year ago	2.263	+3.3%
3 years ago	2.370	-1.3%
5 years ago	2.226	+5.0%

Reflects closing levels to 8-Jun-2026 (ratio 2.338). The spread is up vs every horizon except 3 years, confirming the recent persistent elevation.

10.3 Ratio distribution statistics — full history vs recent 6 years

Statistic	Full history (2007–26)	Recent 6 years
Mean	2.097	2.206
Standard deviation	0.287	0.104
Minimum	1.215	1.893
Maximum	2.665	2.436
Median	2.136	2.214
10th percentile	1.692	2.073
90th percentile	2.450	2.337
Current percentile	81%	90%

The current reading of 2.338 sits at the 81st percentile on 19-year history but the 96th on the recent 6-year window — the crux of the lookback-sensitivity caveat.

Data & methodology. Daily closing levels of NSE ^NSEBANK and ^NSEI, 17-Sep-2007 to 8-Jun-2026 (source: Yahoo Finance), n = 4590 sessions. Ratio, z-scores, base rates and the backtest were computed in Python; costs modelled at 6 bps per turnover unit. Statistical relationships are descriptive and regime-dependent; they do not guarantee future outcomes. **This document is a quantitative strategy note prepared for discussion. It is not investment advice and not a solicitation to trade. The author is not a registered investment adviser.**